BML Munjal University
A Hero GROUP INITIATIVE

Himank Swami

B.Tech – Computer Science and Engineering — 2024 – 2028

Education

B.Tech, Computer Science and Engineering — 2024 – 2028
CGPA: 7.69 / 10.0

Class XII – Science — 68%
Class X — 88%

Research Interests

LLM Inference and Quantization, Agentic RAG Architectures, Multi-modal Vision-Language Models, Medical Image Segmentation, AI for Oncology, Model Ensembling, Safety Evaluation of Large Language Models

Technical Skills

Programming: Python, Java, C, Bash
Machine Learning & Vision: PyTorch, nnU-Net, YOLO, U-Net, DenseNet, Transformers, **Generative AI & LLMs:** Ollama, Qwen, Llama, RAG, RVC (Voice Conversion)
Scientific Computing: NumPy, Pandas, Matplotlib
Web & Deployment: Streamlit, Gradio
Tools: Linux, Git, MySQL, ChromaDB (Vector DB)
Core Concepts: Model Ensembling, Transfer Learning, 4-bit Quantization, ML Deployment Pipelines, Evaluation Framework Design

Projects

AI Brain Tumor Segmentation Platform

Tech: nnU-Net, PyTorch, Gradio, Three.js

- Developed 5-fold ensemble nnU-Net architecture for volumetric MRI brain tumor segmentation.
- Implemented reproducible inference pipelines for medical imaging workflows.
- Generated high-fidelity 3D mesh reconstructions for surgical visualization.
- Designed automated radiological summarization module.

Medical LLM Inference Benchmark — GitHub

- Built specialized evaluation suite for open-source medical LLMs.
- Implemented 4-bit quantized inference for efficient evaluation of 7B+ parameter models.
- Designed multi-dimensional scoring framework (ethics, hallucination, factuality, simplification, summarization).
- Conducted adversarial safety stress testing and latency profiling.

Neuro-Clone (Personal AI Entity)

Tech: Qwen 3.5, Ollama, ChromaDB, RVC (Voice Conversion), Linux, Python

- Developed a local multimodal AI agent using Qwen 3.5 4B and Ollama, optimized for real-time inference on an RTX 4050 (6GB VRAM).
- Built an Agentic RAG layer using ChromaDB to enable long-term memory and autonomous retrieval of data.

- Optimized system performance by implementing 4-bit quantization (GGUF) and KV-cache management, achieving sub-2s response latency.
- Integrated Retrieval-based Voice Conversion (RVC) and Piper TTS to create a custom, low-latency vocal interface for the AI entity.

EVAS (Emergency Vehicle Assistance System)

Tech: YOLO, Roboflow, Python, Computer Vision

- Developed an object detection system using YOLO architecture to identify ambulances in real-time traffic scenarios.
- Trained and fine-tuned a custom dataset on Roboflow for accurate emergency vehicle detection.
- Implemented preprocessing, annotation, and augmentation pipelines to improve model robustness.
- Designed the system to enable intelligent traffic signal prioritization for emergency vehicles.

Research Experience

Dentilligence *(AI Research Intern)*

- Worked on multi-label oral lesion detection and segmentation.
- Trained and evaluated ensemble CNN architectures for medical imaging tasks.

Aether AI *(AI Research Intern)*

- Developed medical LLM benchmarking and evaluation frameworks.
- Built early-stage volumetric segmentation prototypes.

Achievements

- Elected Vice President, driving campus-wide sustainability initiatives and events.
- Organized and executed 20+ environmental and community-driven events, coordinating logistics, sponsorships, and cross-team operations.
- Student Mentor – Organized 3 workshops with 100+ participants.